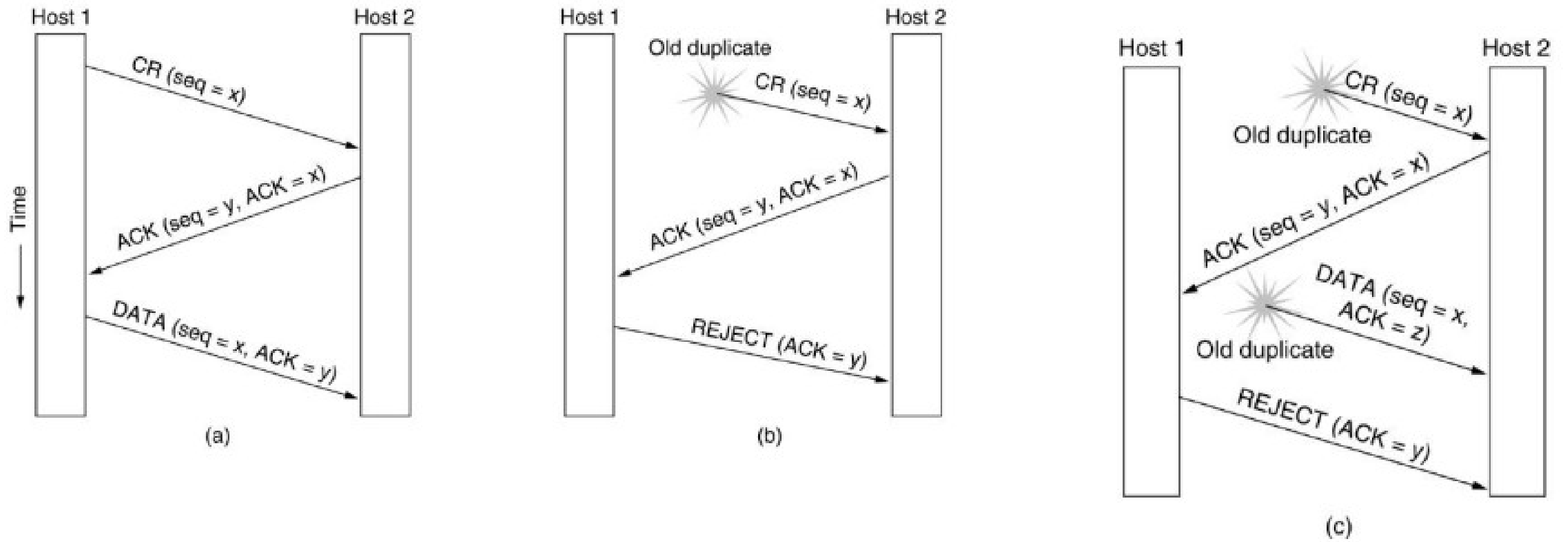
UNIT V

The Internet transport protocol

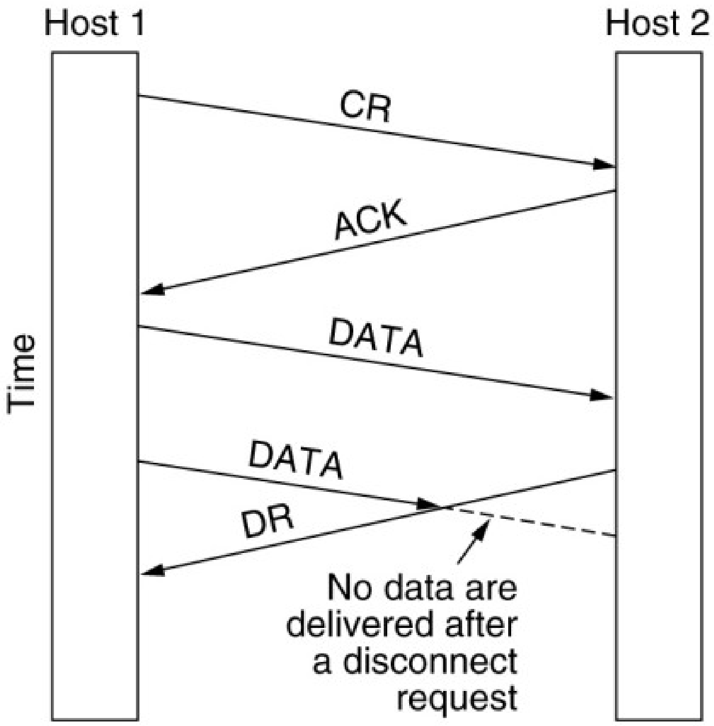
- CONNECTION ESTABLISHMENT Establishing a connection sounds easy, but it is actually surprisingly tricky. At first glance, it would seem sufficient for one transport entity to just send a CONNECTION REQUEST segment to the destination and wait for a CONNECTION ACCEPTED reply. The problem occurs when the network can lose, delay, corrupt, and duplicate packets. This behavior causes serious complications
- To solve this specific problem,(DELAYED DUPLICATES) Tomlinson (1975) introduced the three-way handshake. This establishment protocol involves one peer checking with the other that the connection request is indeed current.
- The normal setup procedure when host 1 initiates is shown in Fig. (a). Host 1 chooses a sequence number, x, and sends a CONNECTION REQUEST segment containing it to host 2. Host 2 replies with an ACK segment acknowledging x and announcing its own initial sequence number, y. Finally, host 1 acknowledges host 2’s choice of an initial sequence number in the first data segment that it sends.

- Connection Establishment



Three protocol scenarios for establishing a connection using a three-way handshake. CR denotes CONNECTION REQUEST.
(a) Normal operation,
(b) Old CONNECTION REQUEST appearing out of nowhere.
(c) Duplicate CONNECTION REQUEST and duplicate ACK.

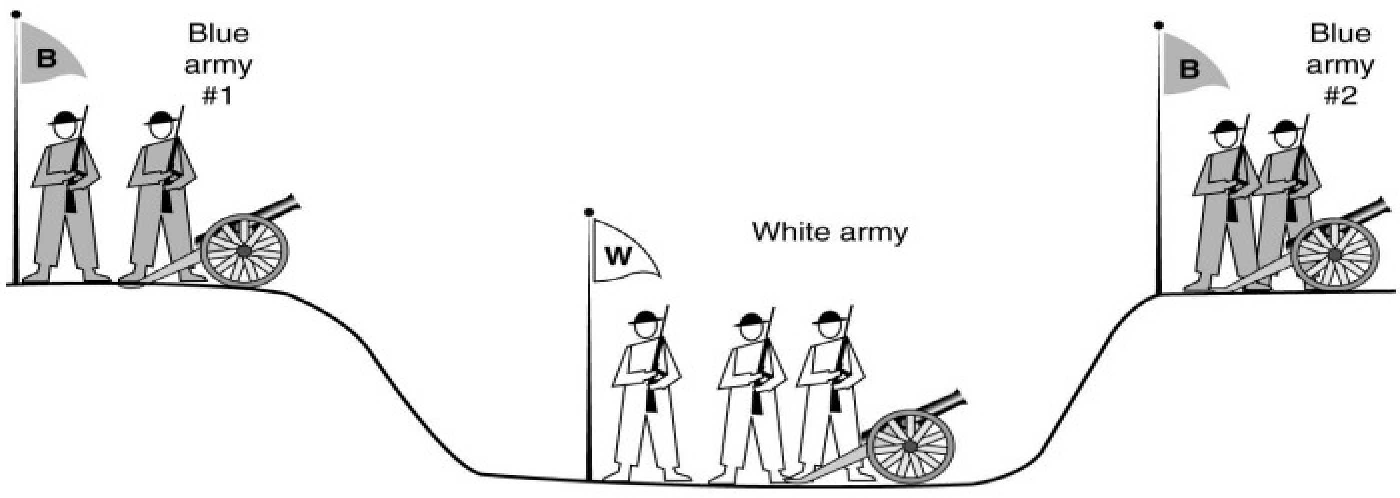
- Connection Release



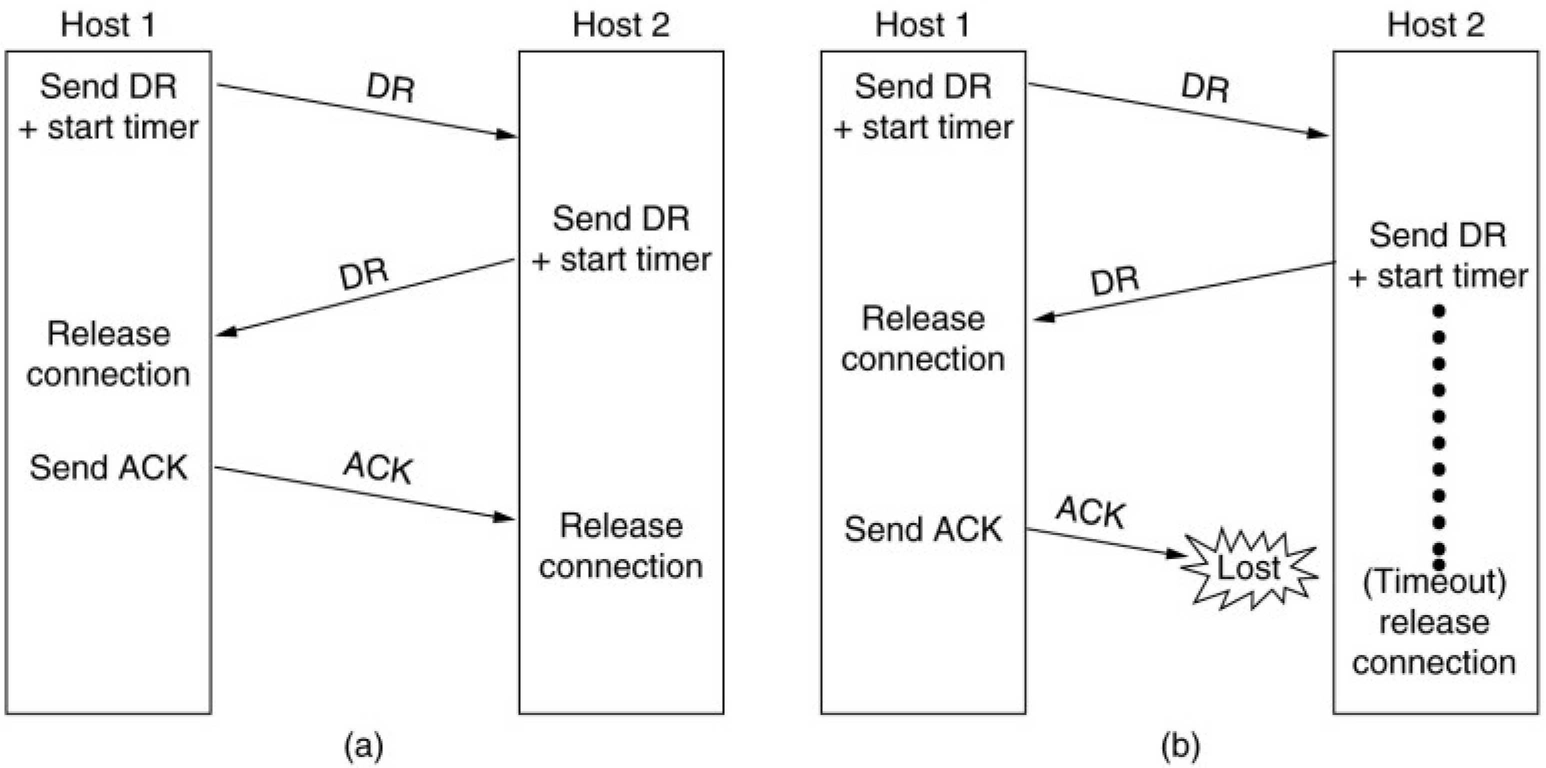
- Abrupt disconnection with loss of data. there are two styles of terminating a connection: asymmetric release and symmetric release Asymmetric release is the way the telephone system works: when one party hangs up, the connection is broken.
- Symmetric release treats the connection as two separate unidirectional connections and requires each one to be released separately Asymmetric release is abrupt and may result in data loss.
- Consider the scenario of Fig. After the connection is established, host 1 sends a segment that arrives properly at host 2. Then host 1 sends another segment. Unfortunately, host 2 issues a DISCONNECT before the second segment arrives. The result is that the connection is released and data are lost.

Connection Release

The two-army problem.

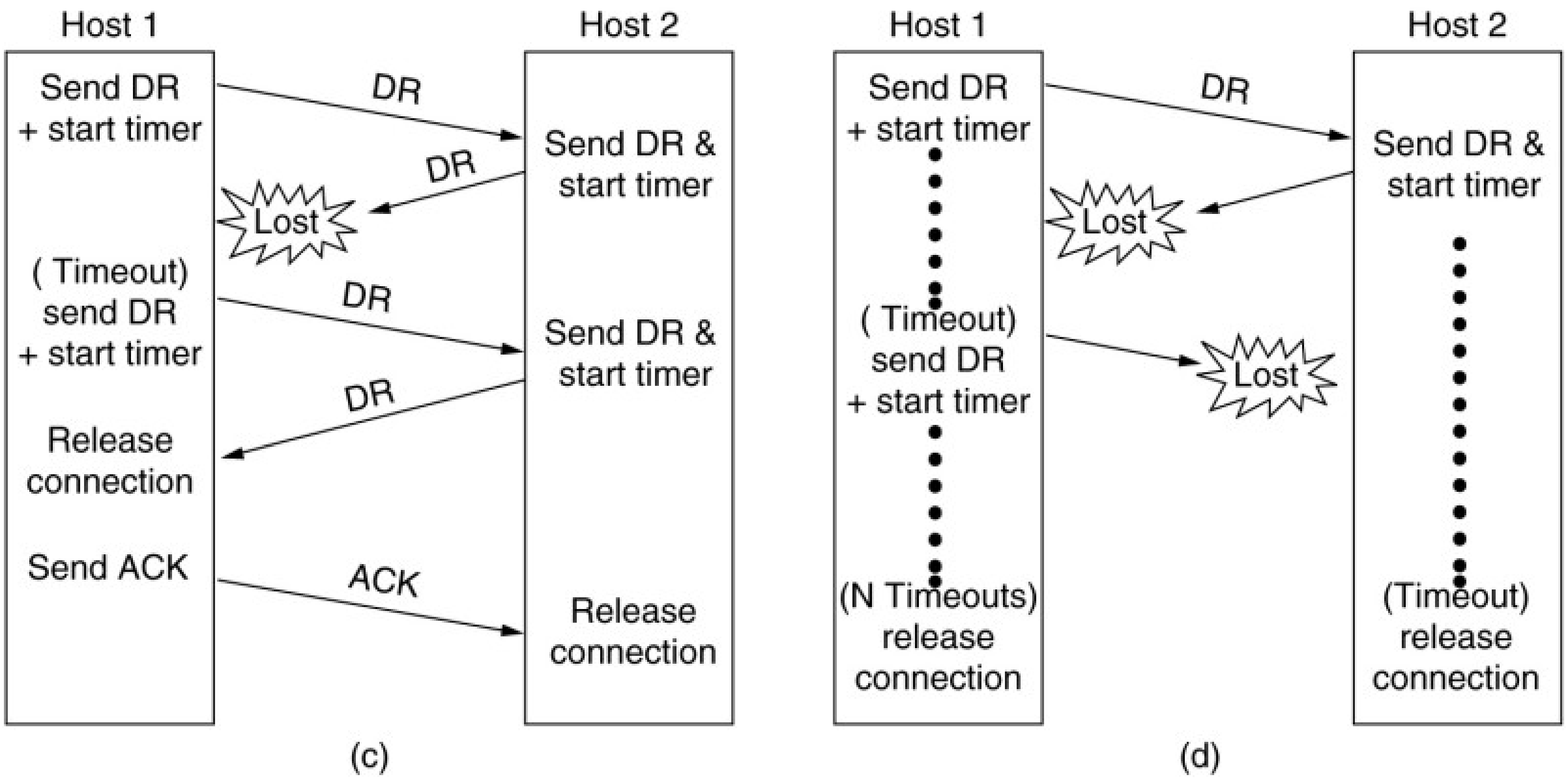


- Connection Release



Four protocol scenarios for releasing a connection. (a) Normal case of a three-way handshake. (b) final ACK lost.

Connection Release



(c) Response lost. (d) Response lost and subsequent DRs lost.

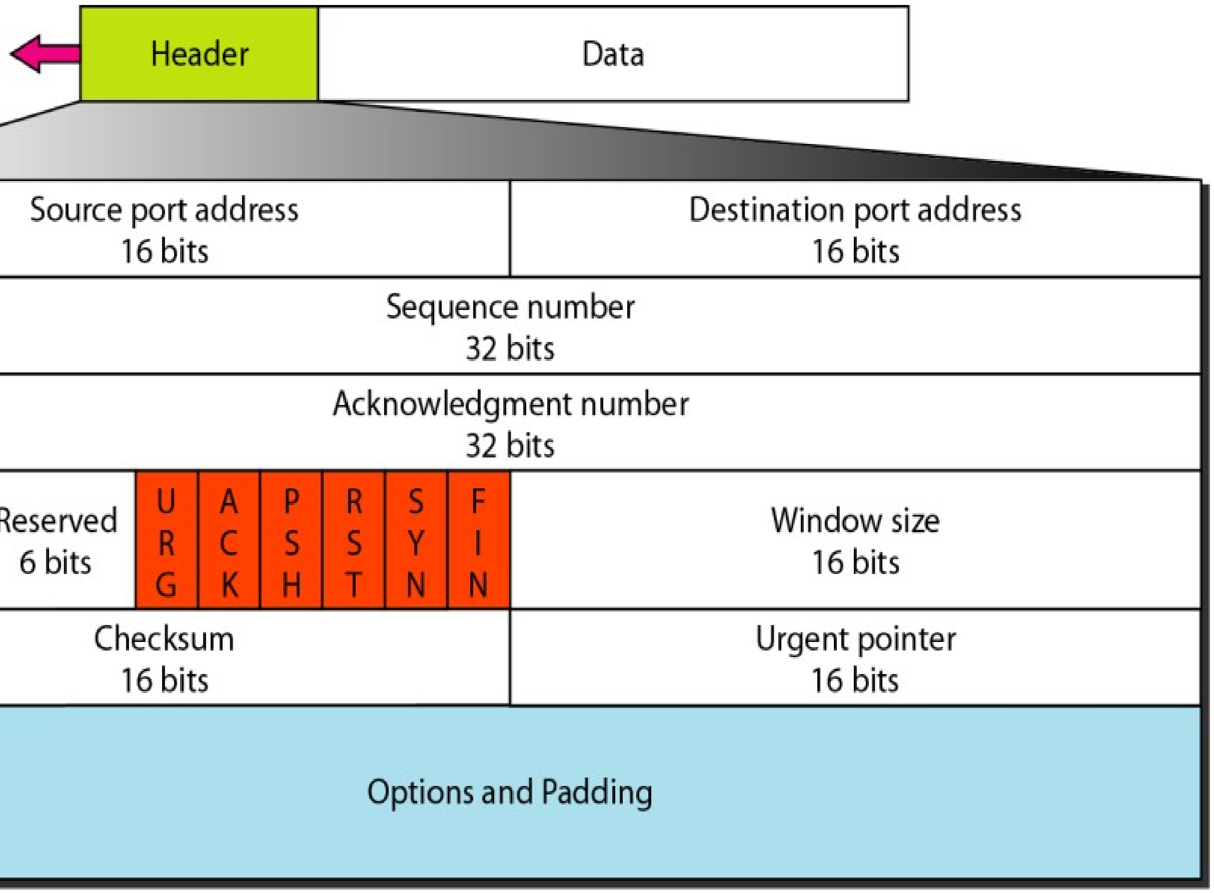
- In Fig. (a), we see the normal case in which one of the users sends a DR (DISCONNECTION REQUEST) segment to initiate the connection release. When it arrives, the recipient sends back a DR segment and starts a timer, just in case its DR is lost. When this DR arrives, the original sender sends back an ACK segment and releases the connection. Finally, when the ACK segment arrives, the receiver also releases the connection. If the final ACK segment is lost, as shown
- In Fig.(b), the situation is saved by the timer. When the timer expires, the connection is released anyway. Now consider the case of the second DR being lost. The user initiating the disconnection will not receive the expected response, will time out, and will start all over again.
- In Fig.(c), we see how this works, assuming that the second time no segments are lost and all segments are delivered correctly and on time. Last scenario,
- Fig.(d), is the same as Fig. (c) except that now we assume all the repeated attempts

  to retransmit the DR also fail due to lost segments.

- TCP Services 1 Process-to-Process Communication TCP provides process-to-process communication using port numbers. Below Table lists some well-known port numbers used by TCP.

Port	*Protocol*	*Description*
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
20	FTP, Data	File Transfer Protocol (data connection)
21	FTP, Control	File Transfer Protocol (control connection)
23	TELNET	Terminal Network
25	SMTP	Simple Mail Transfer Protocol
53	DNS	Domain Name Server
67	BOOTP	Bootstrap Protocol
79	Finger	Finger
80	HTTP	Hypertext Transfer Protocol
111	RPC	Remote Procedure Call

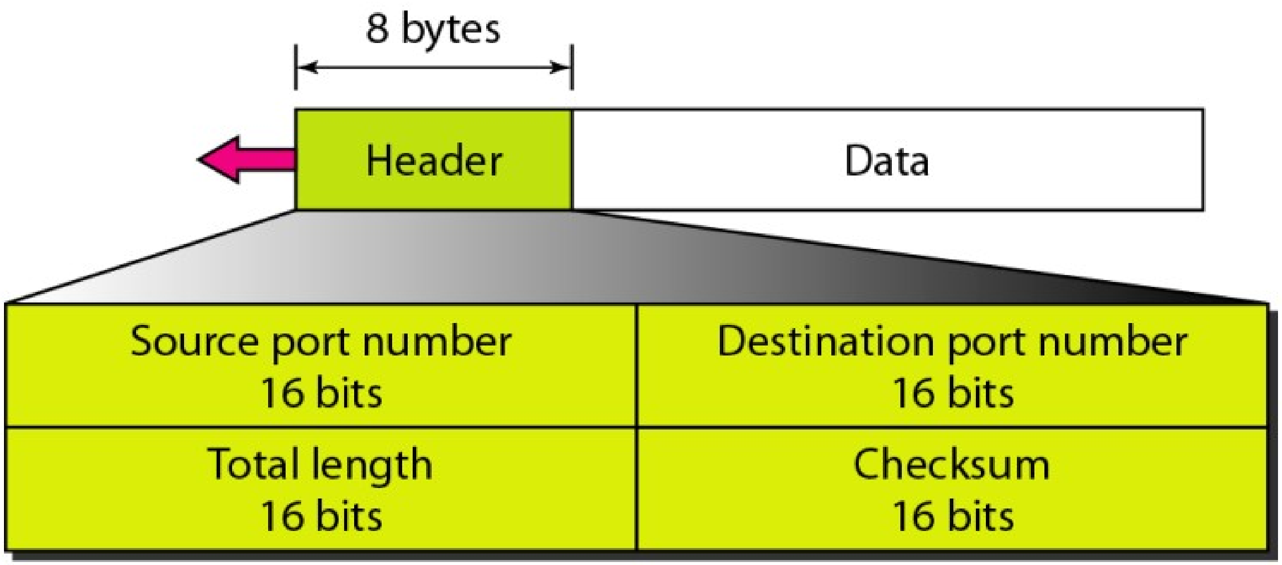
- **TCP segment format**



- The segment consists of a 20- to 60-byte header,.
- Source port address. This is a 16-bit field that defines the port number of the application program in the host that is sending the segment.
- Destination port address. This is a 16-bit field that defines the port number of the application program in the host that is receiving the segment.
- Sequence number. This 32-bit field defines the number assigned to the first byte of data contained in this segment. As we said before, TCP is a stream transport protocol. To ensure connectivity, each byte to be transmitted is numbered. The sequence number tells the destination which byte in this sequence comprises the first byte in the segment. During connection establishment, each party uses a random number generator to create an initial sequence number (ISN), which is usually different in each direction.
- Acknowledgment number. This 32-bit field defines the byte number that the receiver of the segment is expecting to receive from the other party. If the receiver of the segment has successfully received byte number x from the other party, it defines x + I as the acknowledgment number. Acknowledgment and data can be piggybacked together.
- Header length. This 4-bit field indicates the number of 4-byte words in the TCP header. The length of the header can be between 20 and 60 bytes. Therefore, the value of this field can be between 5 (5 x 4 =20) and 15 (15 x 4 =60).
- Reserved. This is a 6-bit field reserved for future use. Control. This field defines 6 different control bits or flags as shown in Figure. One or more of these bits can be set at a time.

- These bits enable flow control, connection establishment and termination, connection abortion, and the mode of data transfer in TCP.
- Window size. This field defines the size of the window, in bytes, that the other party must maintain. Note that the length of this field is 16 bits, which means that the maximum size of the window is 65,535 bytes. This value is normally referred to as the receiving window (rwnd) and is determined by the receiver. The sender must obey the dictation of the receiver in this case.
- Checksum. This 16-bit field contains the checksum. The calculation of the checksum for TCP follows the same procedure as the one described for UDP. However, the inclusion of the checksum in the UDP datagram is optional, whereas the inclusion of the checksum for TCP is mandatory. The same pseudoheader, serving the same purpose, is added to the segment. For the TCP pseudoheader, the value for the protocol field is 6.
- Urgent pointer. This l6-bit field, which is valid only if the urgent flag is set, is used when the segment contains urgent data. It defines the number that must be added to the sequence number to obtain the number of the last urgent byte in the data section of the segment. This will be discussed later in this chapter. Options. There can be up to 40 bytes of optional information in the TCP header. We will not discuss these options here; please refer to the reference list for more information.

- USER DATAGRAM PROTOCOL (UDP) The User Datagram Protocol (UDP) is called a connectionless, unreliable transport protocol. It does not add anything to the services of IP except to provide process-to-process communication instead of host-tohost communication.



- Checksum (OPTIONAL,IF NOT USED SET ALL 1'S DEFAULT) The UDP checksum calculation is different from the one for IP and ICMP. Here the checksum includes three sections: a pseudo header, the UDP header, and the data coming from the application layer.
- The pseudo header is the part of the header of the IP packet in which the user datagram is to be encapsulated with some fields filled with Os If the checksum does not include the pseudo header, a user datagram may arrive safe and sound.
- However, if the IP header is corrupted, it may be delivered to the wrong host. The protocol field is added to ensure that the packet belongs to UDP, and not to other transport-layer protocols.

- Remote Procedure Call The key work was done by Birrell and Nelson (1984). In a nutshell, what Birrell and Nelson suggested was allowing programs to call procedures located on remote hosts. When a process on machine 1 calls a procedure on machine 2, the calling process on 1 is suspended and execution of the called procedure takes place on 2.
- Information can be transported from the caller to the callee in the parameters and can come back in the procedure result. No message passing is visible to the application programmer. This technique is known as RPC (Remote Procedure Call). Traditionally, the calling procedure is known as the client and the called procedure is known as the server, and we will use those names here too.

- Problems with RPC: 1 With RPC, passing pointers is impossible because the client and server are in different address spaces. 2 It is essentially impossible for the client stub to marshal the parameters: it has no way of determining how large they are. 3 A third problem is that it is not always possible to deduce the types of the parameters, not even from a formal specification or the code itself.(exa: printf) 4 A fourth problem relates to the use of global variables. Normally, the calling and called procedure can communicate by using global variables, in addition to communicating via parameters. But if the called procedure is moved to a remote machine, the code will fail because the global variables are no longer shared.

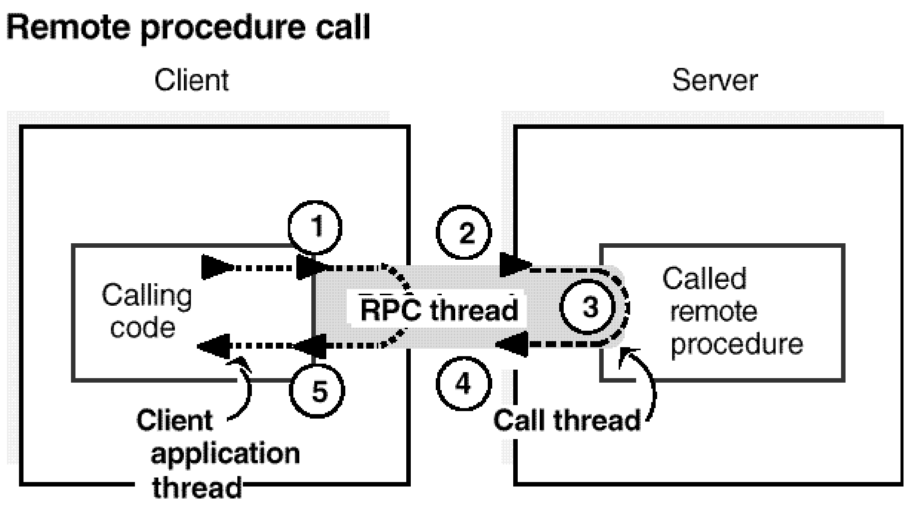
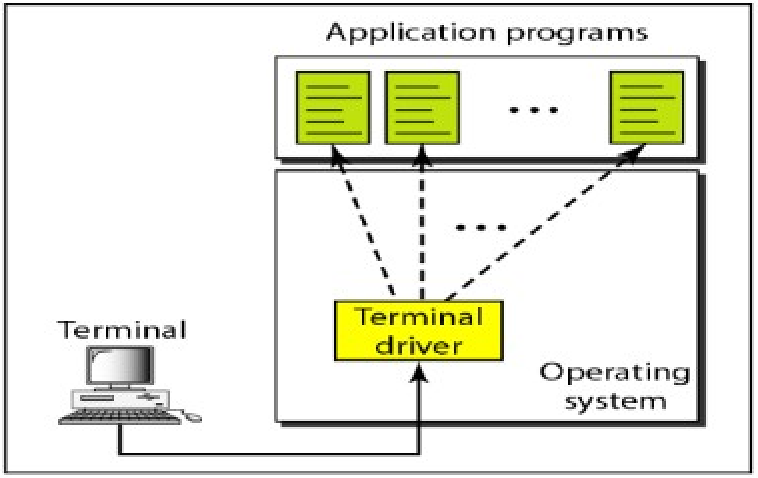


Figure 6-1 Execution Phases of an RPC Thread

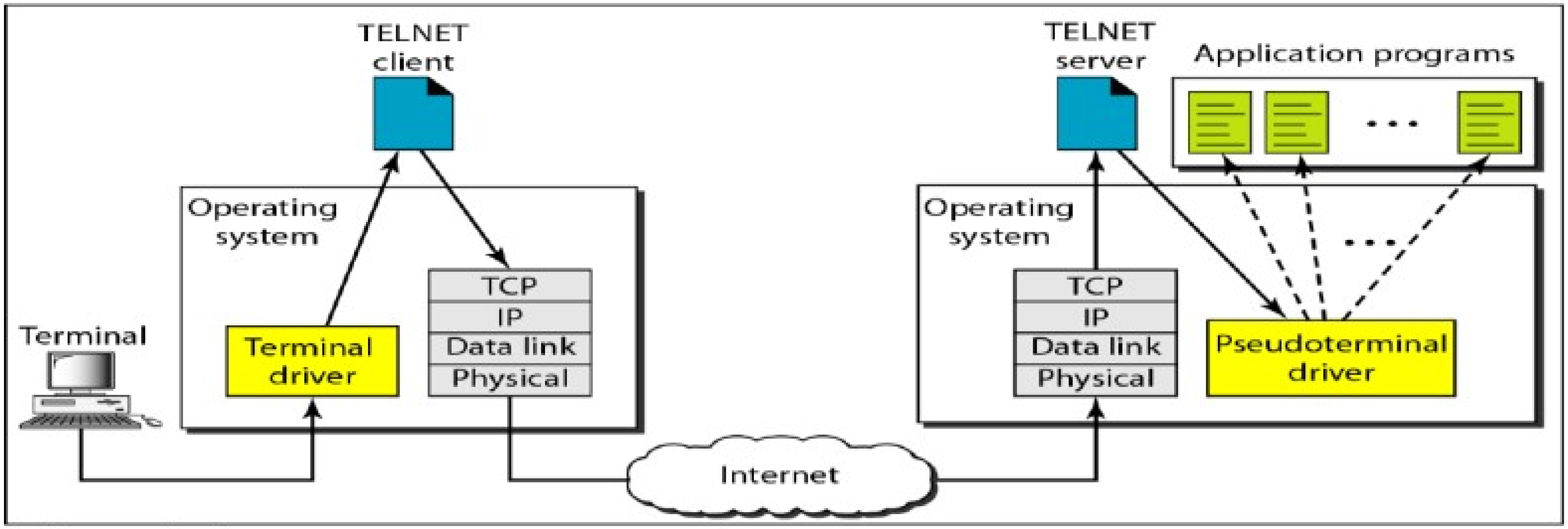
Figure: Execution Phases of an RPC Thread

- TELNET It is client/server application program. TELNET is an abbreviation for TErminaL NETwork. TELNET enables the establishment of a connection to a remote system in such a way that the local terminal appears to be a terminal at the remote system. Timesharing Environment A large computer supports multiple users.
- The interaction between a user and the computer occurs through a terminal, which is usually a combination of keyboard, monitor, and mouse. Logging To access the system, the user logs into the system with a user id or log-in name.
- The system also includes password checking to prevent an unauthorized user from accessing the resources. Local login Remote login.

- TELNET



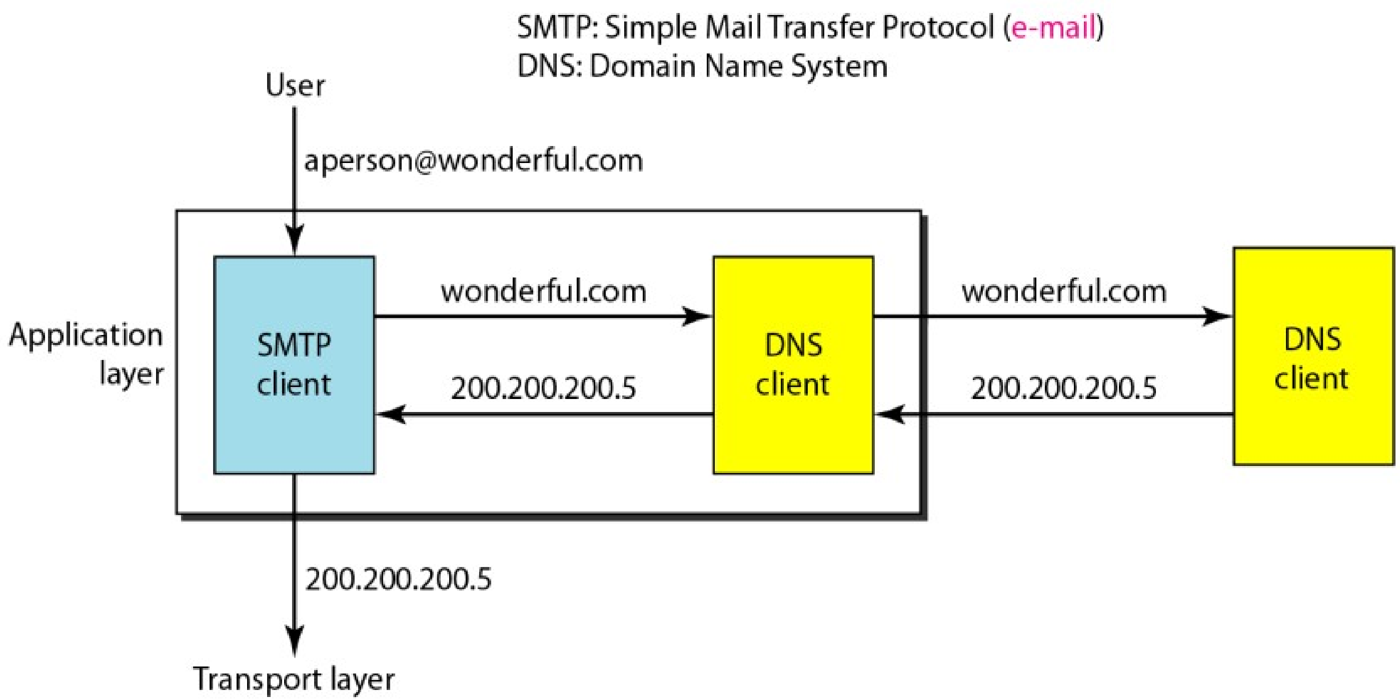
a. Local log-in



b. Remote log-in

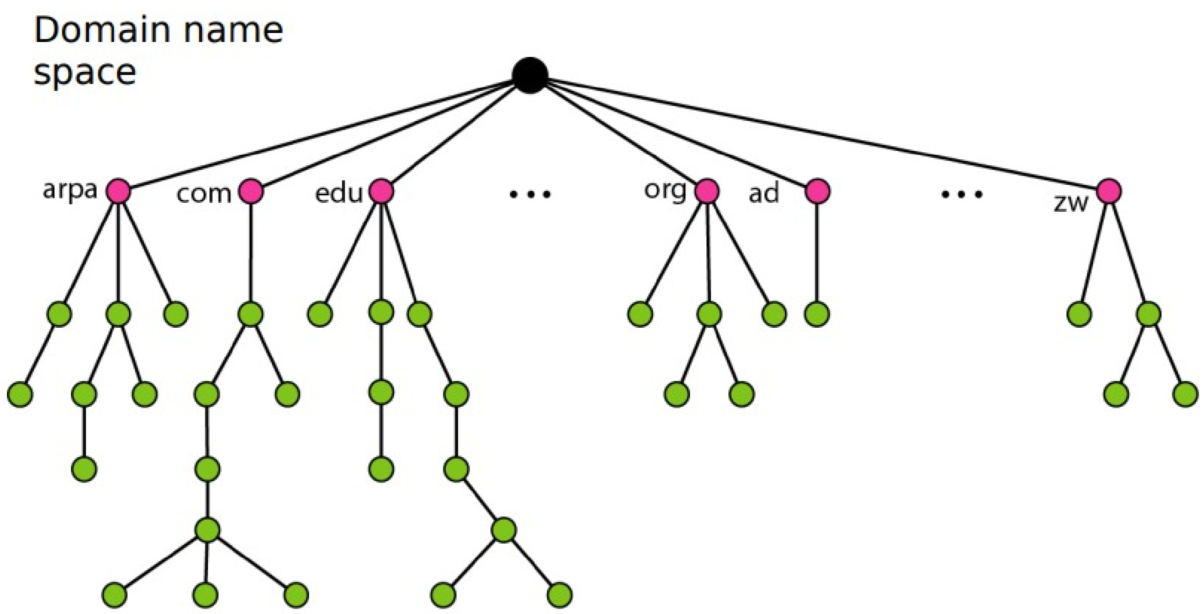
- When a user logs into a local timesharing system, it is called local log-in. As a user types at a terminal or at a workstation running a terminal emulator, the keystrokes are accepted by the terminal driver. The terminal driver passes the characters to the operating system.
- The operating system, in turn, interprets the combination of characters and invokes the desired application program or utility. When a user wants to access an application program or utility located on a remote Machine, it is called remote log-in. Here the TELNET client and server programs come into use.
- The user sends the keystrokes to the terminal driver, where the local operating system accepts the characters but does not interpret them. The characters are sent to the TELNET client, which transforms the characters to a universal character set called network virtual terminal (NVT) characters and delivers them to the local TCP/IP protocol stack.

- DNS (Domain Name System) To identify an entity, TCP/IP protocols use the IP address, which uniquely identifies the connection of a host to the Internet. However, people prefer to use names instead of numeric addresses. Therefore, we need a system that can map a name to an address or an address to a name.

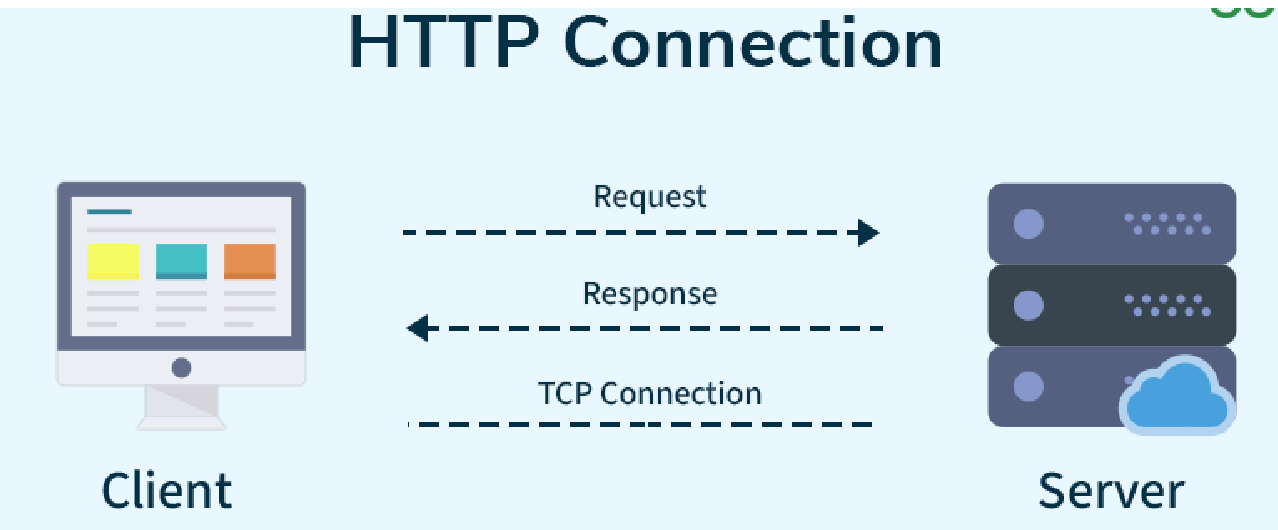


- NAME SPACE A name space that maps each address to a unique name can be organized in two ways: fiat or hierarchical.
- Flat Name Space In a flat name space, a name is assigned to an address. A name in this space is a sequence of characters without structure.
- Hierarchical Name Space In a hierarchical name space, each name is made of several parts. The first part can define the nature of the organization, the second part can define the name of an organization, the third part can define departments in the organization, and so on. Exa: challenger.jhda.edu, challenger.berkeley.edu, and challenger.smart.com
- DOMAIN NAME SPACE To have a hierarchical name space, a domain name space was designed. In this design the names are defined in an inverted-tree structure with the root at the top. The tree can have only 128 levels: level 0 (root) to level 127.

- Label Each node in the tree has a label, which is a string with a maximum of 63 characters. The root label is a null string (empty string).
- Domain Name Each node in the tree has a domain name. A full domain name is a sequence of labels separated by dots (.). The domain names are always read from the node up to the root.

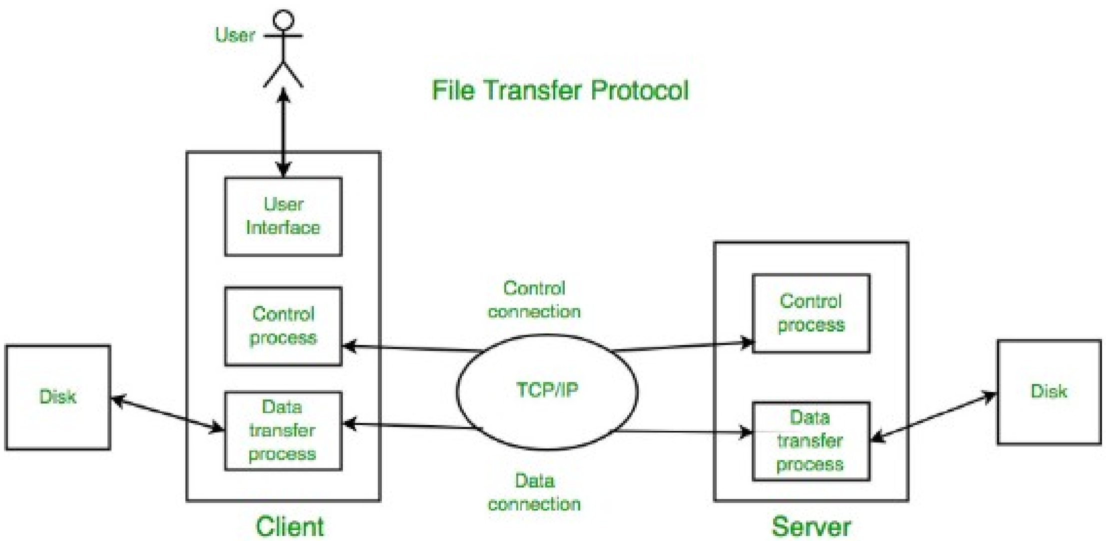


- **HTTP** stands for **Hypertext Transfer Protocol**, and it's the system that allows communication between web browsers (like Google Chrome or Firefox) and websites. When you visit a website, your browser uses HTTP to send a request to the server hosting that site, and the server sends back the data needed to display the page.
- Think of HTTP as a set of rules or a language used by your browser and the web server to talk to each other, ensuring that websites load properly when you type in their URLs



- **How HTTP Works: Step-by-Step Process**
- Here's how HTTP works when you visit a website:
- **Open Web Browser**: First, you open your web browser and type a website URL (e.g., www.example.com).
- **DNS Lookup**: Your browser asks a Domain Name System (DNS) server to find out the IP address associated with that URL. Think of this as looking up the phone number of the website.
- **Send HTTP Request**: Once the browser has the website's IP address, it sends an HTTP **request** to the server. The request asks the server for the resources needed to display the page (like text, images, and videos).
- **Server Response**: The server processes your request and sends back an HTTP **response**. This response contains the requested resources (like HTML, CSS, JavaScript) needed to load the page.
- **Rendering the Web Page**: Your browser receives the data from the server and displays the webpage on your screen.

- File transfer protocol (FTP) is an Internet tool provided by TCP/IP. It helps to transfer files from one computer to another by providing access to directories or folders on remote computers and allows software, data and text files to be transferred between different kinds of computers.
- It encourages the direct use of remote computers.
- It shields users from system variations (operating system, directory structures, file structures, etc.)
- It promotes the sharing of files and other types of data.



- **Types of Connection in FTP**
- Control Connection
- Data Connection
- **Control Connection**
- For sending control information like user identification, password, commands to change the remote directory, commands to retrieve and store files, etc., FTP makes use of a control connection. The control connection is initiated on port number 21.
- **Data connection**
- For sending the actual file, FTP makes use of a data connection. A data connection is initiated on port number 20.
  FTP sends the control information out-of-band as it uses a separate control connection. Some protocols send their request and response header lines and the data in the same TCP connection. For this reason, they are said to send their control information in-band. HTTP and SMTP are such examples.

- **Electronic Mail** (e-mail) is one of most widely used services of Internet. This service allows an Internet user to send a **message in formatted manner (mail)** to the other Internet user in any part of world. Message in mail not only contain text, but it also contains images, audio and videos data. The person who is sending mail is called **sender** and person who receives mail is called **recipient**. It is just like postal mail service. **Components of E-Mail System :** The basic components of an email system are : User Agent (UA), Message Transfer Agent (MTA), Mail Box, and Spool file. These are explained as following below.
- **User Agent (UA) :** The UA is normally a program which is used to send and receive mail. Sometimes, it is called as mail reader. It accepts variety of commands for composing, receiving and replying to messages as well as for manipulation of the mailboxes.
- **Message Transfer Agent (MTA) :** MTA is actually responsible for transfer of mail from one system to another. To send a mail, a system must have client MTA and system MTA. It transfer mail to mailboxes of recipients if they are connected in the same machine. It delivers mail to peer MTA if destination mailbox is in another machine. The delivery from one MTA to another MTA is done by Simple Mail Transfer Protocol.

Components of E-mail System

1.User Agent(UA)

It is a program that is mainly used to send and receive an email. It is also known as an email reader. User-Agent is used to compose, send and receive emails.

It is the first component of an Email.

User-agent also handles the mailboxes.

The User-agent mainly provides the services to the user in order to make the sending and receiving process of message easier.

2.Message Transfer Agent

The actual process of transferring the email is done through the Message Transfer Agent(MTA).

In order to send an Email, a system must have an MTA client.

In order to receive an email, a system must have an MTA server.

The protocol that is mainly used to define the MTA client and MTA server on the internet is called SMTP(Simple Mail Transfer Protocol).

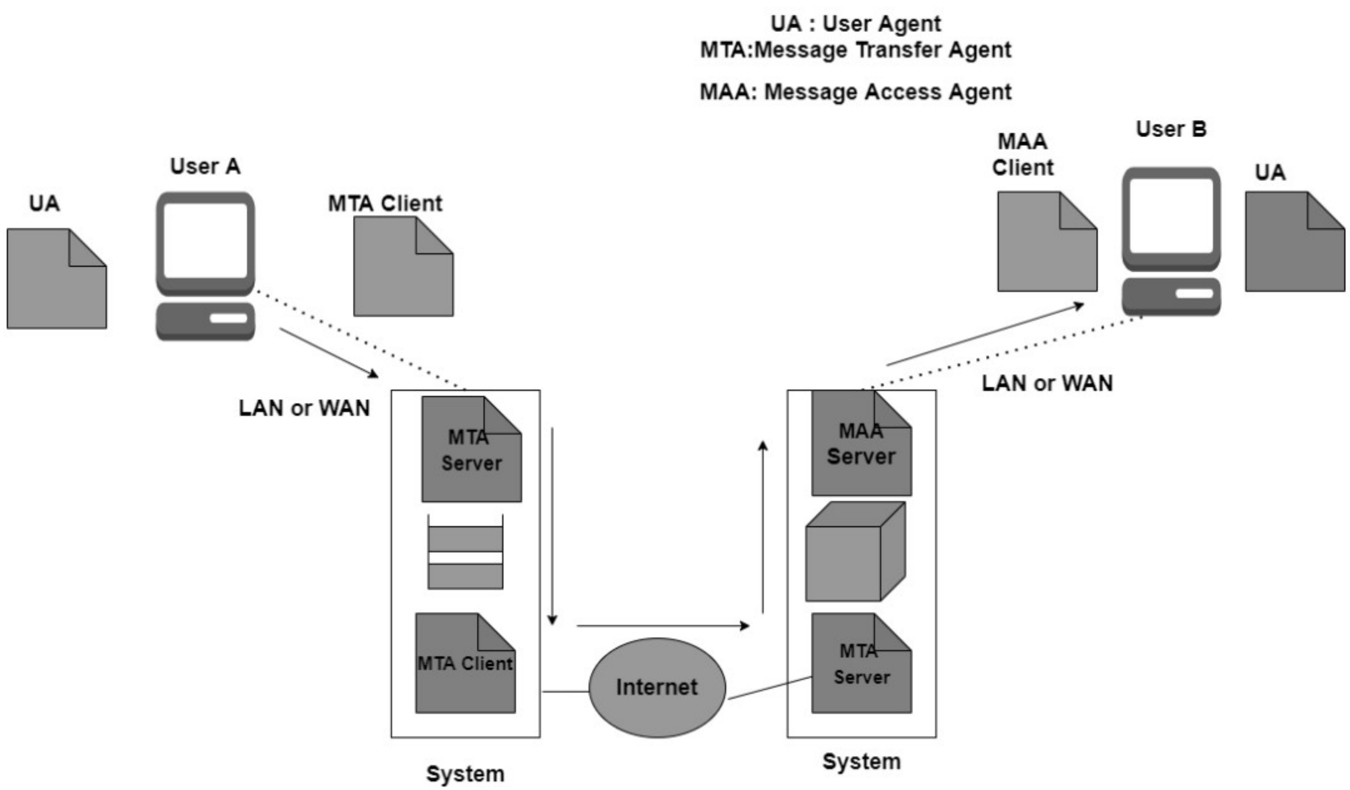
3.Message Access Agent

In the first and second stages of email delivery, we make use of SMTP.

SMTP is basically a Push protocol.

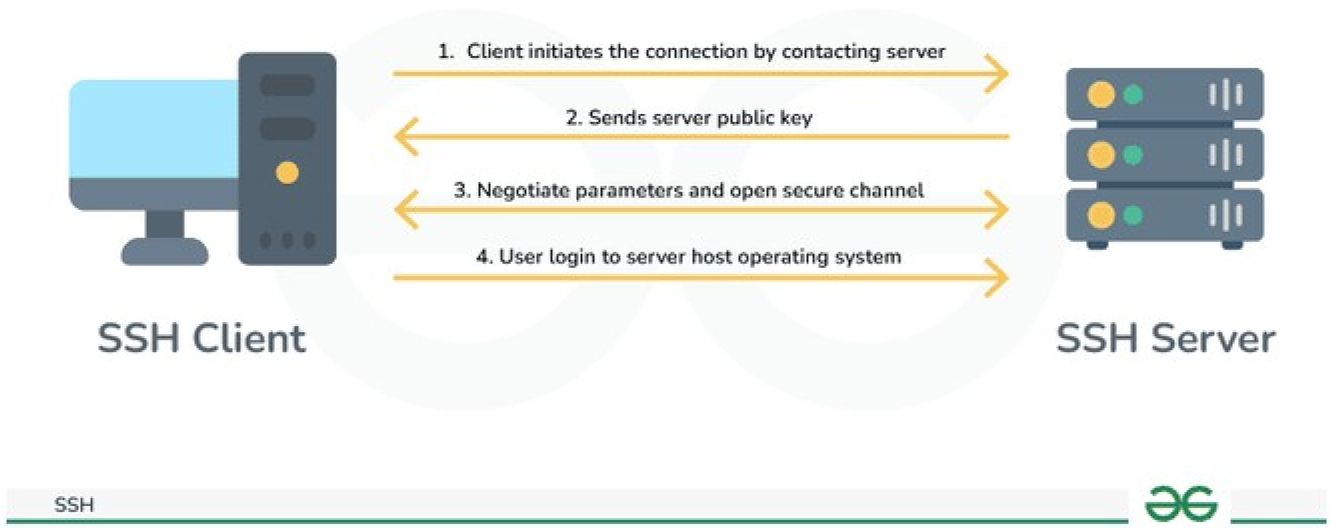
The third stage of the email delivery mainly needs the pull protocol, and at this stage, the message access agent is used.

- **Mailbox :** It is a file on local hard drive to collect mails. Delivered mails are present in this file. The user can read it delete it according to his/her requirement. To use e-mail system each user must have a mailbox . Access to mailbox is only to owner of mailbox.
- **Spool file :** This file contains mails that are to be sent. User agent appends outgoing mails in this file using SMTP. MTA extracts pending mail from spool file for their delivery. E-mail allows one name, an **alias**, to represent several different e-mail addresses.



- The **SSH (Secure Shell)** is an access credential that is used in the SSH Protocol. In other words, it is a cryptographic network protocol that is used for transferring encrypted data over the network. The port number of SSH is 22. It allow users to connect with server, without having to remember or enter password for each system. It always comes in key pairs:
- **Public key -** Everyone can see it, no need to protect it. (for encryption function).
- **Private key -** Stays in computer, must be protected. (for decryption function).
- Secure Shell or SSH, is a protocol that allows you to connect securely to another computer over an unsecured network. It developed in 1995. SSH was designed to replace older meth
- Imagine a system administrator working from home who needs to manage a remote server at a company data center. Without SSH, they would have to worry about their login credentials being intercepted, leaving the server vulnerable to hackers. Instead of it after using SSH, the administrator establishes a secure connection that encrypts all data sent over the internet. They can now log in with their username and a private key, allowing them to safely execute commands on the server, transfer files, and make necessary updates, all of these without the risk of spying eyes

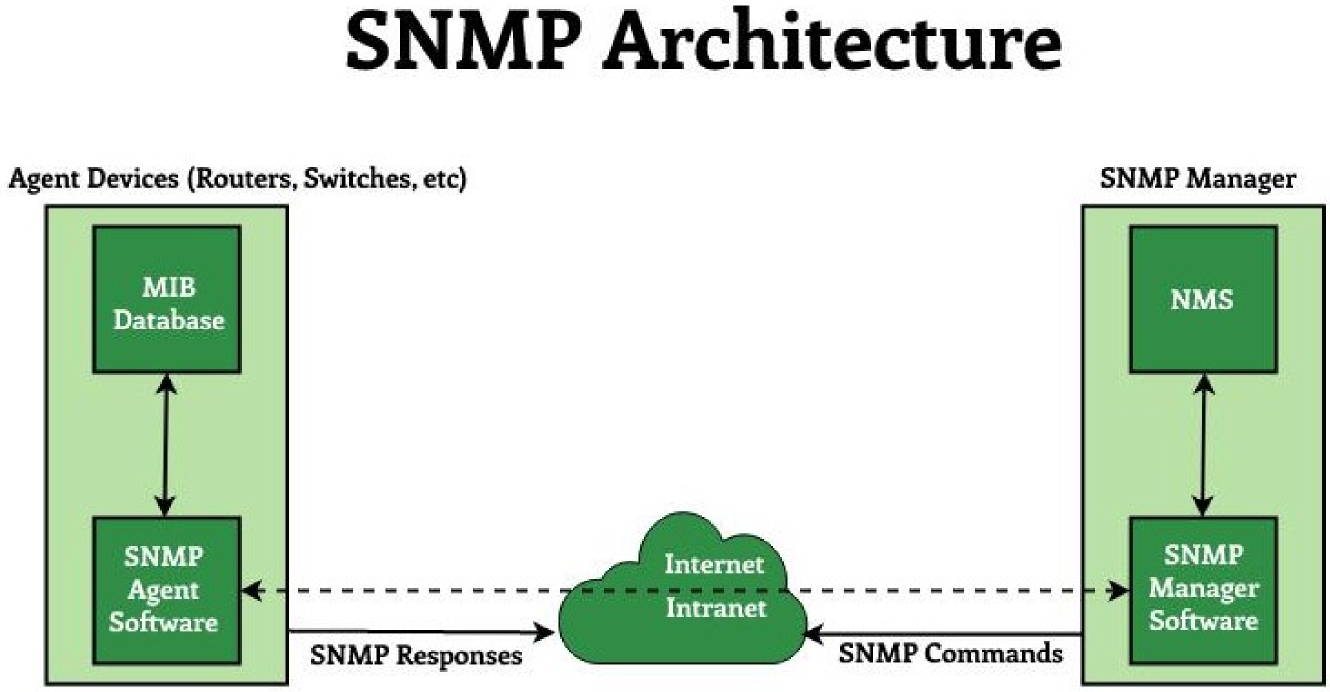
- **SSH Functions**
- There are multiple functions performed by SSH Function, here below are some functions:
- SSH provides high security as it encrypts all messages of communication between client and server.
- SSH provides confidentiality.
- SSH allows remote login, hence is a better alternative to TELNET.
- SSH provides a secure File Transfer Protocol, which means we can transfer files over the Internet securely.
- SSH supports tunneling which provides more secure connection communication.



SSH Protocol

- **Simple Network Management Protocol (SNMP)** is an Internet Standard protocol used for managing and monitoring network-connected devices in IP networks. SNMP is an application layer protocol that uses UDP port number 161/162. SNMP is used to monitor the network, detect network faults, and sometimes even to configure remote devices.
- **Architecture of SNMP**
- There are mainly three main components in SNMP architecture:
- **SNMP Manager:** It is a centralized system used to monitor the network. It is also known as a Network Management Station (NMS). A router that runs the SNMP server program is called an agent, while a host that runs the SNMP client program is called a manager.
- **SNMP agent:** It is a software management software module installed on a managed device. The manager accesses the values stored in the database, whereas the agent maintains the information in the database. To ascertain if the router is congested or not, for instance, a manager can examine the relevant variables that a router stores, such as the quantity of packets received and transmitted.

- **Management Information Base:**MIB consists of information on resources that are to be managed. This information is organized hierarchically. It consists of objects instances which are essentially variables. A MIB, or collection of all the objects under management by the manager, is unique to each agent. System, interface, address translation, IP, UDP , and EGP , ICMP , TCP are the eight categories that make up MIB. The MIB object is home to these groups.



- World Wide Web (WWW), often called the Web, is a system of interconnected webpages and information that you can access using the Internet. It was created to help people share and find information easily, using links that connect different pages together. The Web allows us to browse websites, watch videos, shop online, and connect with others around the world through our computers and phones.
- All public websites or web pages that people may access on their local computers and other devices through the internet are collectively known as the World Wide Web or W3. Users can get further information by navigating to links interconnecting these pages and documents. This data may be presented in text, picture, audio, or video formats on the internet.
- **Key Parts of the Web**
- The Web has three main building blocks that make it work:
- **URL (Uniform Resource Locator**: This is the address of a webpage, like https://www.example.com./ It tells your browser exactly where to find the page.
- **HTTP (Hypertext Transfer Protocol)**: This is the set of rules that lets your browser and the server talk to each other to send and receive webpages.
- **HTML (Hypertext Markup Language)**: This is the code that tells browsers how to display a webpage, including where to put text, pictures, and links.